BarSync

MIDI CLOCK BAR COUNTER & VISUALIZER

Quickstart Guide

Always in Time.

OVERVIEW

What is BarSync?

BarSync keeps you on top of your track's arrangement — live or in the studio. It receives your sequencer's MIDI clock and counts and visualizes the bars elapsed since a starting point you define yourself — either the first MIDI clock signal, or a manual reset via button. That way you always know exactly which bar you're in, and drops, breaks and builds never catch you off guard.

Features

- Bar/beat counter, synced to the incoming MIDI clock (24 PPQN)
- Grid progress bar (x1 to x128 bars) — footprint stays constant, tile size adapts automatically
- 5 selectable time signatures (3/4, 4/4, 5/4, 6/8, 7/8)
- Reset button with two stages (bar end / cycle end), plus a standalone SET 1.1 function to instantly lay down a new beatgrid (quantized or instant)
- MIDI Monitor built into the device (Note/CC/PC/Pitch-Bend log, MIDI clock and RUN/STOP graph)
- Nudge mode to manually compensate for clock phase drift
- MIDI Thru to pass the signal on to further devices
- Standby (light sleep) on MIDI inactivity
- Full settings menu right on the device — no app, no software needed

Connection & First Start

1. **Connect your MIDI source.**
   Connect your sequencer's/DAW interface's MIDI OUT to BarSync's **MIDI IN**.

2. **Optional: pass it on.**
   MIDI THRU lets you forward the incoming signal unchanged to another device — handy when BarSync sits in the middle of an existing MIDI chain.

3. **Power it up.**
   BarSync starts automatically and shows the retro-style boot screen for about 5 seconds.

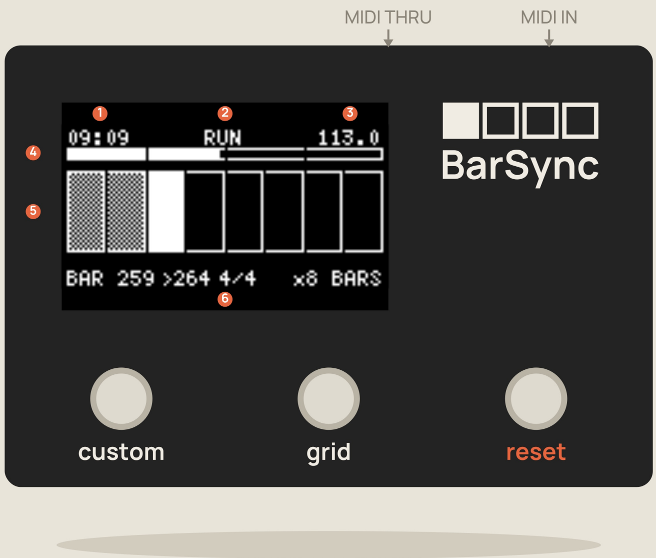
4. **Start the MIDI clock.**
   As soon as your sequencer is running, the status switches from **STOP** to **RUN**, and the bar/beat counter starts running.

5. **Set a reference point (optional).**
   The 1.1 is set automatically as soon as BarSync receives your sequencer's **Start/Run signal** — you don't have to do anything for that. If you'd rather start counting from a specific point instead, just press the **Reset** button.

> If no MIDI clock arrives for a while, the display shows **"NO MIDI-CLOCK"** instead — a cue to check your MIDI cable and source (see Troubleshooting on the last page).

OPERATION

The Three Buttons



The Display at a Glance

NO.	SHOWS
1	Play time (mm:ss) since the last start/reset
2	Status — RUN or STOP
3	Tempo in BPM
4	Progress within the current bar
5	Progress within the current grid cycle — one tile = one bar
6	current bar, end of cycle, time signature, grid size

custom

- **Short press:** SET 1.1 — instantly lays down a new beatgrid (default assignment).
- **Hold 1 second:** toggle MIDI Monitor on/off.
- **Together with Grid:** toggle Nudge mode on/off.

grid

- **Short press:** cycles the number of bars shown in the grid (x1 to x128).

reset

- **Short press (Reset 1):** waits for the end of the current bar.
- **Hold (Reset 2):** waits for the end of the current grid cycle.
- **Hold 1 second while booting:** opens the settings menu.
- **On the MIDI Monitor screen:** clears the note/CC list instead.

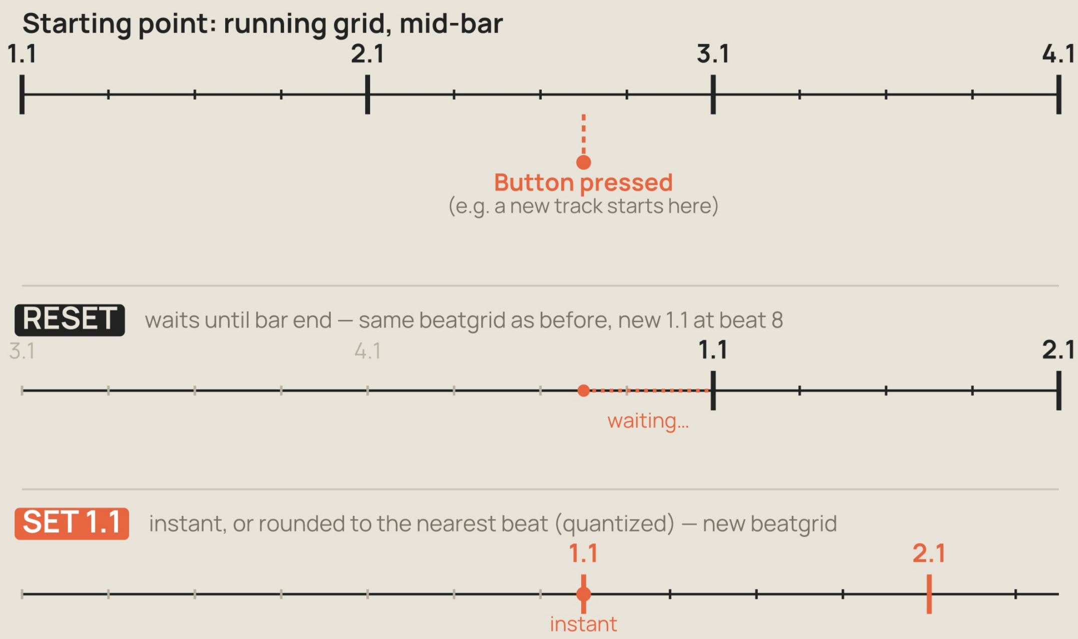
The Custom button is freely assignable — the functions shown here match the firmware default (Switches › Custom › Function).

Reset vs. SET 1.1 — What's the Difference?

The 1.1 is set all by itself to begin with: as soon as BarSync receives your sequencer's **Start/Run signal**, counting starts automatically at 1.1 — without you pressing any button.

You only need **SET 1.1** when you want to correct that automatically-set point afterwards — for example when you've started a new song during your live set and your track's 1.1 no longer "sits" on BarSync's 1.1. Sure, you could restart the sequencer and everything would line up again. But during a performance that's obviously something you'd rather avoid. That's where SET 1.1 comes in: pressing the **Custom** button lays down a new beatgrid with 1.1 exactly at that moment.

The two **Reset** stages (1 and 2) build directly on that beatgrid and simply reset the bar counter including the grid (and, depending on settings, the play time) — always in sync with the current beatgrid.



FUNCTION	EFFECT
Reset	Waits for the end of the current bar (Reset 1) or cycle (Reset 2) and then resets to 1.1. The **beatgrid stays unchanged** — the new bar boundaries land exactly where a bar boundary already was.
SET 1.1	Lays down a **completely new beatgrid** instantly (Instant) or rounded to the nearest beat (Quantized) — regardless of where the previous beatgrid sat. As a result, every following bar boundary can shift compared to before.

> Rule of thumb: if your set runs continuously within the same beatgrid, **Reset** is all you need. Only once a new song section starts with its own downbeat that doesn't match your current count do you need **SET 1.1** to re-anchor the beatgrid.

The Settings Menu

Open it: hold the Reset button for 1 second while powering on.

SETUP

- TIMESIG — time signature, direct value (3/4, 4/4, 5/4, 6/8, 7/8), default 4/4
- SWITCHES ›
  - CUSTOM › Function (Timesig / SET 1.1) — depending on Function, underneath: time-signature selection or Mode (Quantized/Instant)
  - GRID › choice of which divisor steps are selectable
  - RESET › playtime reset per stage (Yes/No)
- DISPLAY ›
  - CONTRAST, INVERT (swap black/white), GRID LINES (separators every 32 bars, toggleable)
- STANDBY — on/off, sleep delay
- DEFAULTS — load factory settings (with confirmation)

Navigating the Menu

BUTTON	FUNCTION IN THE MENU
Custom	move up one line
Grid	move down one line
Reset (short)	open selection / change value
Reset (hold 1s)	back one level
Reset (on page 1, hold 3s)	save, exit menu, restart

MIDI Monitor

The built-in MIDI Monitor shows you, right on the device, what's actually arriving on the MIDI line — handy for debugging your setup without any separate software or a computer sitting next to you.

Show/Hide

Hold the Custom button for 1 second (default assignment).

What's Shown?

- Event log of the last 6 Note On/Note Off, CC, Program Change and Pitch Bend messages, including channel — newest message on top
- MIDI clock graph: one spike per detected beat, the downbeat (1.1) drawn twice as wide as a regular beat
- MIDI RUN/STOP trace as a level graph, with a plain-text "RUN = HIGH" / "RUN = LOW" status

> On the Monitor screen, the **Reset** button takes on a different job: it clears the note/CC list instead of triggering the normal reset function.

> If no MIDI clock arrives for more than 5 seconds, "NO CLOCK" blinks in place of the clock graph instead.

REFERENCE

Troubleshooting & Quick Reference

PROBLEM	SOLUTION
Display stays dark	Check power supply; check contrast in the setup menu (Display › Contrast)
"NO MIDI-CLOCK" is shown	Check the MIDI cable and source; make sure the clock is actually running
Buttons don't respond / respond incorrectly	Restart the device; if in doubt, load factory settings (Setup › Defaults)
Device only responds after a delay following a long pause	Normal behavior: BarSync goes into standby (light sleep) on MIDI inactivity and wakes automatically on new activity

Quick Reference

BUTTON	FUNCTION
Custom (short)	SET 1.1 — lay down a new beatgrid*
Custom (hold 1s)	toggle MIDI Monitor on/off*
Grid (short)	cycle the number of bars shown (x1–x128)
Reset (short/hold)	Reset stage 1/2
Custom + Grid	toggle Nudge mode on/off*
Hold Reset while booting	settings menu

*The Custom button is freely assignable — this reflects the firmware default.

Full documentation, schematic, firmware and 3D-printable enclosure:
github.com/barna81/BarSync